

# **SBML Level 3 Package: Flux Balance Constraints version 3**

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## **Abstract**

Constraint-based modeling is a well-established modeling methodology used to study biological networks at both the medium-scale and genome-scale. Due to their large size and complexity, such steady-state flux models are typically analyzed using constraint-based optimization techniques, such as Flux Balance Analysis (FBA).

The Flux Balance Constraints (FBC) Package extends SBML Level 3 to provide a standardized format for encoding, exchanging, and annotating constraint-based models. It includes support for modeling concepts such as objective functions, flux bounds, and annotation of model components that facilitate reaction balancing. Version two extended the original release by adding support for encoding gene-protein associations.

Version three builds upon and maintains backwards compatibility with Version two by introducing new elements and attributes that include: user-defined constraints, a new quadratic variable type and key-value pair annotation that can be used to store additional information relevant to constraint-based modeling. The only changes to existing attributes are that a Species charge is no longer restricted to an integer value and chemicalFormula can now include generic components.

In addition to providing the elements necessary to uniquely encode existing constraint-based models, the FBC package provides an open platform that facilitates the continued, cross-community development of an interoperable, constraint-based model encoding format.